MIL-R-25536A(USAF)
AMENDMENT 1
26 June 1959

MILITARY SPECIFICATION

ROCKET MOTORS, AERONAUTICAL, ACCEPTANCE TEST FOR

This amendment forms a part of Military Specification MIL-R-25536A(USAF) dated 6 April 1959.

Page 2, Paragraph 3.1.1.2.1: In the second line delete "free" and substitute "ambient".

Page 2, Paragraph 3.1.1.3: In the fifth line delete "any" and substitute "all".

Page 6, Paragraph 4.2: Delete and substitute

"4.2 Acceptance Test.- All rocket motors submitted for acceptance shall be subjected to the acceptance tests specified herein. Acceptance of production rocket motors shall be by sublot for sequential sampling plans and by individual rocket motors for regression analysis plans."

Page 6, Paragraph 4.2.1: Delete and substitute

"4.2.1 Inspection Groups"

Page 6, Add the following paragraphs

"4.2.1.1 Lot.- A lot of rocket motors shall consist of a number of rocket motors produced in a continuous production period with no change in process specifications or in raw material lots.

"4.2.1.1.1 Inspection Lot.- The inspection lot for nondestructive inspection, within the limitations of 4.2.1.1, shall be designated by the procuring activity and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

"4.2.1.1.2 Acceptance Analysis Lot.- The acceptance analysis lot shall consist of all rocket motors fired from ten consecutive acceptable sublots.

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"4.2.1.2 Sublot

"4.2.1.2.1 Production Sublot.- A sublot of production rocket motors shall consist of either of the following:

a. Not less than ten and not more than twenty rocket motors manufactured from the same propellant batch.

b. Ten rocket motors manufactured from two or more propellant batches, where individual rocket motors may contain propellant from one batch only or from more than one batch.

"4.2.1.2.2 Preliminary Flight Rating and Qualification Test Sublot.- A sublot of preliminary flight rating test or qualification test rocket motors shall consist of five rocket motors that meet all nondestructive inspection requirements."

Page 7, Paragraph 4.2.2.1.3: In the second line delete "static" and substitute "hydrostatic".

Page 7, Paragraph 4.2.2.1.4: In the second line delete "static" and substitute "hydrostatic".

Page 8, Paragraph 4.2.3.2.1: Delete and substitute

"4.2.3.2.1 Sampling Plans.- The acceptance test sampling plans for firing tests shall be designed by the contractor in accordance with the requirements herein and submitted to the procuring activity for approval as part of the detailed acceptance test method. "

Page 8, Add the following paragraphs

"4.2.3.2.1.1 Preliminary Flight Rating and Qualification Test Plan.- Rocket motors built for use in the preliminary flight rating test or qualification test shall be accepted thru use of a plan based upon analysis by variables in accordance with the requirements of MIL-STD-414 using the tightest AQL for the sample size involved to determine maximum allowable percent defective. One rocket motor shall be selected at random from each sublot. This rocket motor shall be static tested and the test data combined with the results of the last four previous acceptable firings, making the cumulative sample size five. (Data from development firings of rocket motors of the final configuration shall be used until sufficient firings have been obtained from preliminary flight rating or qualification test acceptance.) If variables analysis of the data from the five tests shows the rocket motors to be acceptable, the rocket motors shall be accepted for use in the preliminary flight rating or qualification

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test. If variables analysis of the data from the five tests shows the rocket motors to be unacceptable, the four untested rocket motors from which the last sample fired was selected shall be rejected and the test data eliminated from the cumulative sample data.

*4.2.3.2.1.2 Production Acceptance Plan.- Rocket motors built for use in the production quality test or built for production acceptance shall be accepted thru use of a production test plan of one of the types specified below. After the qualification test has been approved, the data from all rocket motors fired (including rocket motors of the qualified configuration that were fired during development) shall be used to arrive at values for standard deviation and mean for all rocket motor ballistic parameters. The values of standard deviation and mean thus obtained for each ballistic parameter shall be utilized in the production acceptance plan.

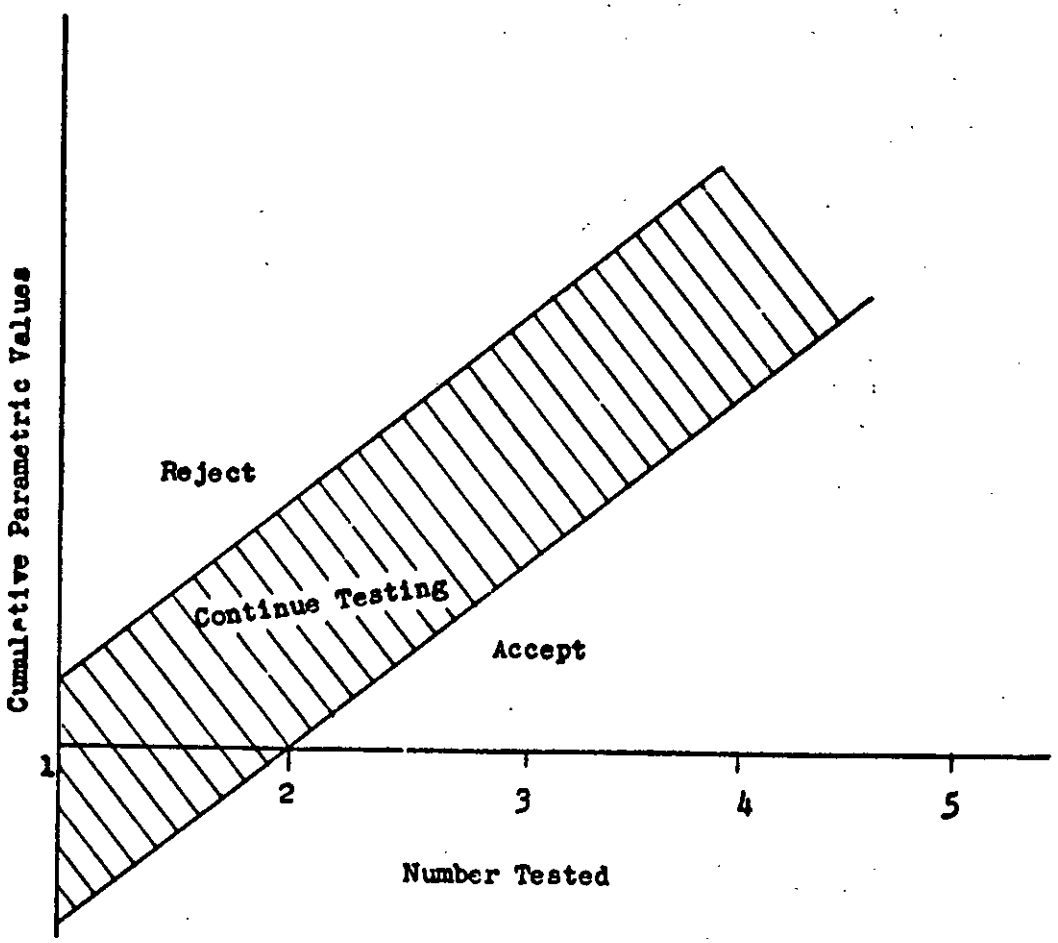
*4.2.3.2.1.2.1 Sequential Sampling.- Any sequential sampling plan shall be based upon the following:

a. The probability of acceptance when the percent defective is 20 percent or more shall be 0.01.

b. The probability of rejection when the percent defective is 12 percent or less shall be 0.05.

Acceptance and rejection limits shall be calculated for each parameter. Accept-reject limits for each parameter shall be included in graphical form. (These graphs shall be drawn in accordance with figure 2.) No more than 50 percent of the rocket motors from any sublot shall be fired to determine acceptability of the sublot. For any sublot, all firings shall be at the same test temperature. If the sublot is not accepted after the maximum permissible number of firings has been made, the data from these firings shall be subjected to an analysis by variables in accordance with the procedures of MIL-STD-414 using the tightest AQL for the number of firings involved. If the predicted percent defective is equal to or less than that for the sample size and AQL involved, the remaining rocket motors of the sublot shall be accepted. If the predicted percent defective is greater than that for the sample size and AQL involved, the remaining rocket motors of the sublot shall be rejected and the data excluded from computations of mean and standard deviation. If three sublots are rejected from any ten consecutive sublots submitted for acceptance, no additional sublots shall be manufactured for acceptance until the rocket motor manufacturer has demonstrated that the cause(s) for rejection has been corrected and the corrective action has been approved by the procuring activity.

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Sequential Sampling Acceptance Curve

Figure 2

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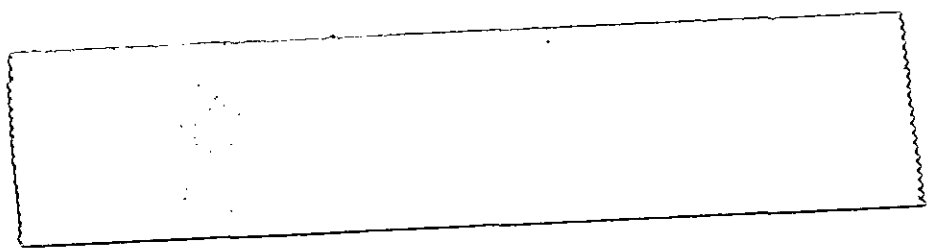
"4.2.3.2.1.2.1.1 Acceptance Analysis Lot.- The test data (grouped by test temperature) for each acceptance analysis lot shall be analyzed by test of significance procedures to determine the drift in population statistics. Testing shall be conducted at the 0.05 level of significance unless otherwise specified by the procuring activity. If the drift is determined to be significant, corrective action shall be taken.

"4.2.3.2.1.2.2 Regression Analysis.- When a regression analysis plan is proposed, the rocket motor development program shall include provisions to establish a correlation between full-scale and small-scale tests. The correlation shall be established prior to qualification testing and shall be demonstrated during the qualification test. If a faulty correlation is disclosed by the qualification test, a revised correlation shall be prepared based upon the qualification test data. The upper and lower limits for the correlation parameters shall be such that rocket motor performance is within the limits of the model specification. At least three small scale tests (one at each test temperature) shall be conducted in conjunction with each full scale rocket motor submitted for acceptance."

Page 8, Paragraph 4.2.3.2.2: Wherever the word "lot(s)" appears in this paragraph, delete the word "lot(s)" and substitute "sublot(s)".

Page 9, Paragraph 4.2.5: Delete "Preproduction Test" and substitute "Production Quality Test", in the paragraph title and in the second line of the paragraph.

Page 9, Paragraph 4.2.6: In the first line delete "of a lot".



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